

SETI Committee Report to IAA Commission 1 - March 2024.

The IAA SETI Committee executive comprises the chair Prof. Michael Garrett (Univ. of Manchester) with co-chairs Profs. Carol Oliver and Andrew Seimion. Lori Walton is the secretary. The last Committee meeting was held during IAC 2023 and this included remote participation of the members via zoom. The chair's notes for this meeting are available at: [removed]

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We were greatly honoured to have Apollo 9 astronaut, Rusty Schweickart, attend IAC 2023 SETI Session A4.1. He also attended the highly successful IAC Breakthrough Listen reception. Kathryn Denning gave another very comprehensive review of Post-detection protocol activities around the world. This was further supplemented by a presentation from Dr. John Elliott on the work of the recently established SETI Post-detection hub at St. Andrews UK - see <https://seti.wp.st-andrews.ac.uk/>. A paper presenting the work of the IAA in this area (Oliver, Denning, Tennen & Garrett) had been presented at SETI session 2. The various presentations and recorded video, are available here as a compressed file: [removed]

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Since IAC 2023 there has been a lot of activity in the area of SETI Post-detection protocols both internal and external to the SETI Committee. The UK SETI Post-detection Hub is very active and a new SETI and Law Working Group has been established by Prof. Michael Bohlander (Univ of Durham) under the auspices of the the International Institute of Space Law (IISL). The IISL initiative goes beyond the traditional themes of post-detection protocols and has three main focus points: (i) General Post-Contact Matters (possibly including METI), (ii) Human ethics baseline for (peaceful) negotiations/relations with ETI and (iii) Hostile Contact – Adaptation of rules of war and re-conceptualisation of planetary defence. The SETI Institute in the US has also indicated an interest in contributing to this topic. The SETI Committee has previously focused on General post-detection protocols and this is unlikely to change. The landscape in this area of our field is significantly more dynamic but also more more complicated with a number of actors active in the field. SETI Committee members are contributing to all of these efforts and we hope to ensure a truly international and scientifically motivated contribution remains part of all the discussions. A Post-detection paper is being prepared (Tennen, Denning, Oliver & Garrett) for presentation at the IAC 2024 SETI Session A4.2.

Over the last 10 years, there has been a major rejuvenation of SETI research around the globe. In relation to this, the Breakthrough Listen initiative is funded for an additional 5 years and is now hosted by the University of Oxford in the UK. SETI research is really becoming a global field with major groups in the US, UK, Australia, South Africa, China, Italy to name only a few countries. We are therefore under pressure to expand the Committee membership but we have

been hesitant to do this until the generic Terms of Reference of IAA Committees is agreed by the IAA BoT. The committee membership currently stands at ~ 100 people. We'd like to expand and retire some of the membership this year.

There continues to be a good deal of publicity surrounding the UAP (Unidentified Anomalous Phenomena) especially with the recent US Congress hearings on the topic. The SETI Committee's focus remains on the search for techno-signatures beyond the Earth's atmosphere.

Various SETI Sessions are scheduled for the IAC 2024 in Milan: SETI Science and Technology (A4.1), SETI & Society (A4.2) plus various interactive presentations. The SETI sessions are scheduled for Tuesday 15th of October in the morning and afternoon - see: <https://iafastro.directory/iac/browse/IAC-24/A4/>

The next SETI Committee meeting will take place during IAC 2024 - probably the day after the SETI Sessions - 16th of October.

Upcoming SETI related meetings include:

Interstellar frontiers SETI, AB and SKA, Perth, Australia, 11-14 March 2024.
First IAA Symposium on Moon Farside Protection to be held in Turin on March 21-22.

Kavli-IAU Symposium (IAUS 387) '(Toward) Discovery Of Life Beyond Earth And Its Impact' (Durham, UK, 14-19 April 2024).

Life in the Universe (LITU), during the XXXII IAU General Assembly (Cape Town, South Africa, 6-15 August 2024).

Italian SETI meeting - before or after IAC 2024?

Mike Garrett, SETI Committee chair.